

A simple empirical model of salt marsh plant spatial distributions with respect to a tidal channel network

Eric W. Sanderson ^{a,*}, Theodore C. Foin ^b, Susan L. Ustin ^a

^a Department of Land, Air and Water Resources, University of California, One Shields Avenue, Davis, CA 95616, USA

^b Department of Agronomy and Range Science, University of California, One Shields Avenue, Davis, CA 95616, USA

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Abstract

Previous work has shown that the distribution of plant species in a salt marsh near Petaluma, CA, is strongly influenced by the location and size of tidal channels. We developed a simple mathematical model to measure the “channel influence” at each point as a cumulative function based on inverse squared distance to channel, length of potentially influential channel, and channel order (a surrogate for channel cross-sectional area.) Plant species distributions predicted with this channel influence function compared favorably to known distributions in Petaluma Marsh. Using this function and probabilities estimated from transect vegetation data, we simulated the two-dimensional pattern of species abundance across the entire site. Comparisons of predictions with independently generated field maps showed comparable patterns of cover for most species. The results indicate that the distribution of vegetation in Petaluma Marsh, both major zones of marsh dominants and patches of minor species, can be described by a single, empirical factor: cumulative distance from tidal channels of different size. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

Numerous studies of salt marsh vegetation have shown strong relationships between zones of plant species distribution and physical gradients related to elevation and tidal flooding (Mahall and Park, 1976a,b,c; Zedler, 1977; Snow and Vince, 1984;

Vince and Snow, 1984; Callaway et al., 1990; Sanchez et al., 1996). It is therefore unsurprising that more recent studies have shown that plant species distributions are also correlated with the location and size of tidal channels, since the tidal channel networks largely control the distribution of tidal flooding within the marsh (Sanderson et al., 1998, 2000).

The areal extent of influence, the species richness of the vegetation, and the identities of the species assemblages have all been shown to vary with channel size and distance from the channel

* Corresponding author. Address: Wildlife Conservation Society, 2300 Southern Blvd., Bronx, NY 10460 USA. Tel.: +1-718-2206825; fax: +1-718-2207114.

E-mail address: esanderson@wcs.org (E.W. Sanderson).

bank in Petaluma Marsh, CA, USA (Sanderson et al., 2000). Similar sequences of species were shown to occur with distance from both smaller and larger channels, but displaced farther away from the larger channels. Although the ecological mechanisms behind these patterns are not understood, these patterns suggest that channel size and distance to the bank are fundamental factors in determining species distributions (Chapman, 1960; Adam, 1990; Sanderson et al., 2000).

Given these observations, we wished to model this system to test its generality using an ecological modelling approach. Ecological models can be used to integrate system components, test overall system understanding and generate novel hypotheses regarding the relationships between system structure and function (Radford and Blackford, 1996; Wu and Levin, 1997; Jørgensen, 1999; Malkina-Pykh, 2000). Moreover spatially explicit models, based in geographic information systems (GIS), can be used to predict spatial distributions of ecological phenomena given underlying spatially explicit drivers (Gillruth et al., 1995; Hill et al., 1997; Ozesmi and Mitsch, 1997; Antoni and Legovi, 1999; Miller and Urban, 1999; Gaff et al., 2000).

The patterns observed by Sanderson et al. (2000) suggested that a simple geographical correlate might be sufficient to predict the broad patterns of plant species distributions in Petaluma Marsh. Moreover, by basing our model in a GIS, it would be possible to extend descriptions of the spatial structure of salt marsh vegetation from one dimensional transect data into spatially complete, two dimensional representations, suitable for comparison to other geographic and remotely sensed data.

In this simulation study we use a common denominator which integrates these aspects of the tidal channel network into an index for prediction of species' spatial locations: cumulative distance to tidal channels. Using a distance based algorithm ignores much of the biological detail about how salt marsh plant species are interacting with their environment and each other in developing a predictive model. However channel distribution and size are relatively easy to map using standard techniques (e.g., global positioning sys-

tems, aerial photography) and distance is easily measured using GIS. By expressing our model as a function of distance, we create a model, which can be applied after relatively simple field measurements to predict plant species distributions given different tidal channel network configurations. Such applications are important to restore and enhance marshes in the San Francisco Bay Estuary.

2. Methods

2.1. Study site

The field observations for this study were made in a tidal salt marsh (Petaluma Marsh, 38° 12' 20" N, 122° 32' 32" W) along the Petaluma River, about 8 km north of the mouth of the river in San Pablo Bay, CA, USA. This marsh is part of a semi-contiguous network of tidal wetlands that stretch along the northern shore of San Pablo Bay and is part of the greater San Francisco Bay Estuary (Josselyn, 1983). Petaluma Marsh is constrained by levees along the Petaluma River to the east and south, and by rising topography on its western margin. Petaluma Marsh was selected for study because of previous studies at the same or nearby locations and the relatively simple structure of the vegetation. (Mahall and Park, 1976a,b,c; Leopold et al., 1993; Zhang et al., 1997; Sanderson et al., 1998, 2000.)

The vegetation of Petaluma Marsh is dominated by *Salicornia virginica* L. (Zhang et al., 1997), a succulent halophyte with extraordinarily high tolerance to saline soils (Flowers et al., 1977). *Spartina foliosa* Trin., a halophytic cordgrass grows in a continuous band below mean high water (Mahall and Park, 1976a) parallel to the Petaluma River and other large channels. *Scirpus robustus* Pursh, a meso-halophytic bulrush grows in large, discontinuous patches between the *Spartina* and *Salicornia*, in association with the largest channels (Ustin et al., 1982; Whitlow, 1982). A suite of other species grow along the banks of smaller channels in the marsh. Typically halophytic species include *Frankenia salina* I.M. Johnston, *Jaumea carnosa* A. Gray, *Grindelia*

stricta DC., *Atriplex patula* L., *Cuscuta salina* Engelm. and *Distichlis spicata* E. Greene. *Achillea millefolium* L. and two invasive species, *Lepidium latifolium* L. and *Rumex crispus* L., were also observed, but are not commonly restricted to salt marsh habitats. Through the remainder of the paper, each species will be referred to by its generic name. All plant names follow the nomenclature of Hickman (1993).

Frankenia, *Grindelia*, *Distichlis*, *Achillea* and *Rumex* have been previously noted in association with mid-sized tidal channels (De Groot, 1927; Purer, 1942; Marshall, 1948; Hinde, 1954; Chapman, 1960; Vogl, 1966; Redfield, 1972; Balling and Resh, 1983; Collins and Resh, 1985; Adam, 1990; Wayne and Parker, 1994, 1996; Sanderson et al., 1998) and are described in detail in Sanderson et al. (2000). *Lepidium* is frequently found interspersed with *Scirpus* near larger channels; *Jaumea* is often found amongst the *Salicornia*, particularly in low lying depressions (Sanderson et al., 2000).

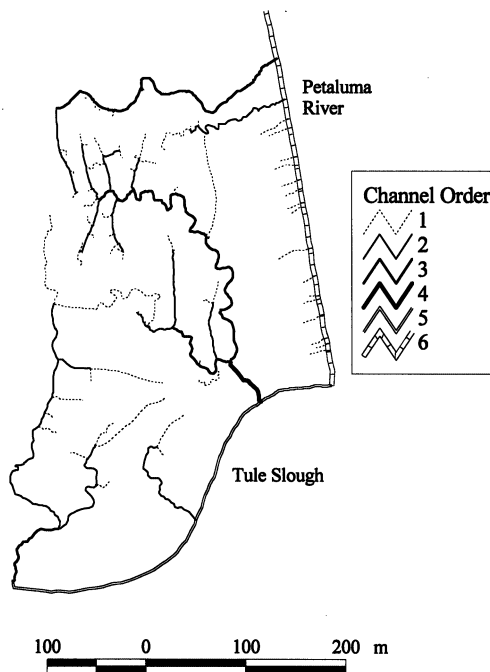


Fig. 1. Map of the modeled tidal channel network in Petaluma Marsh, Sonoma County, CA, USA, with channel order indicated.

A channel network near where Tule Slough (a fifth order channel) empties into the Petaluma River (a sixth order channel) was selected for simulation (Fig. 1). The site covered a rectangular area (roughly 500 m × 300 m², 10.96 ha in area) parallel to the Petaluma River, including stands of *Spartina* and *Scirpus*, as well as large areas of *Salicornia*, and a small channel network. The channel network, containing first – fourth order channels and neighboring portions of Tule Slough and the Petaluma River, was mapped using global positioning systems as described in Sanderson et al. (2000) and the map entered into a GIS. Channel orders, which are a crude but useful measure of channel size in this system, were determined using the method of Strahler (1964).

In Sanderson et al. (2000) we report the results of vegetation transects made perpendicular to tidal channels of different channel order. Each transect consisted of thirty 0.25 m² square quadrats: the first quadrat was laid immediately alongside the channel, the next nineteen quadrats laid every 0.5 meter (i.e., adjacent to the previous quadrat), and the remaining ten quadrats laid every one meter (resulting in thirty observations over twenty meters). For each quadrat, plant cover was estimated according to Braun-Blanquet cover class system (Mueller-Dombois and Ellenberg, 1974.) 900 quadrats were measured at this study site. From these data, average patch size statistics for each species were calculated. Sanderson et al. (2000) for more details.

2.2. Calculation of cumulative inverse squared distance function

Based on the observations reported in Sanderson et al. (2000), we hypothesized that the “channel influence” diminished with distance from the channel bank and that the type and/or quantity of influence varied with channel size. Based on observations on the inside and outside of channel curves, we further hypothesized that the effects of channel geometry appeared to be cumulative. Combining these factors we developed the Cumulative Inverse Squared Distance (CISD) function:

$$\text{Untransformed CISD } (q) = \Sigma(w_0^*d^{-2}), \quad (1)$$

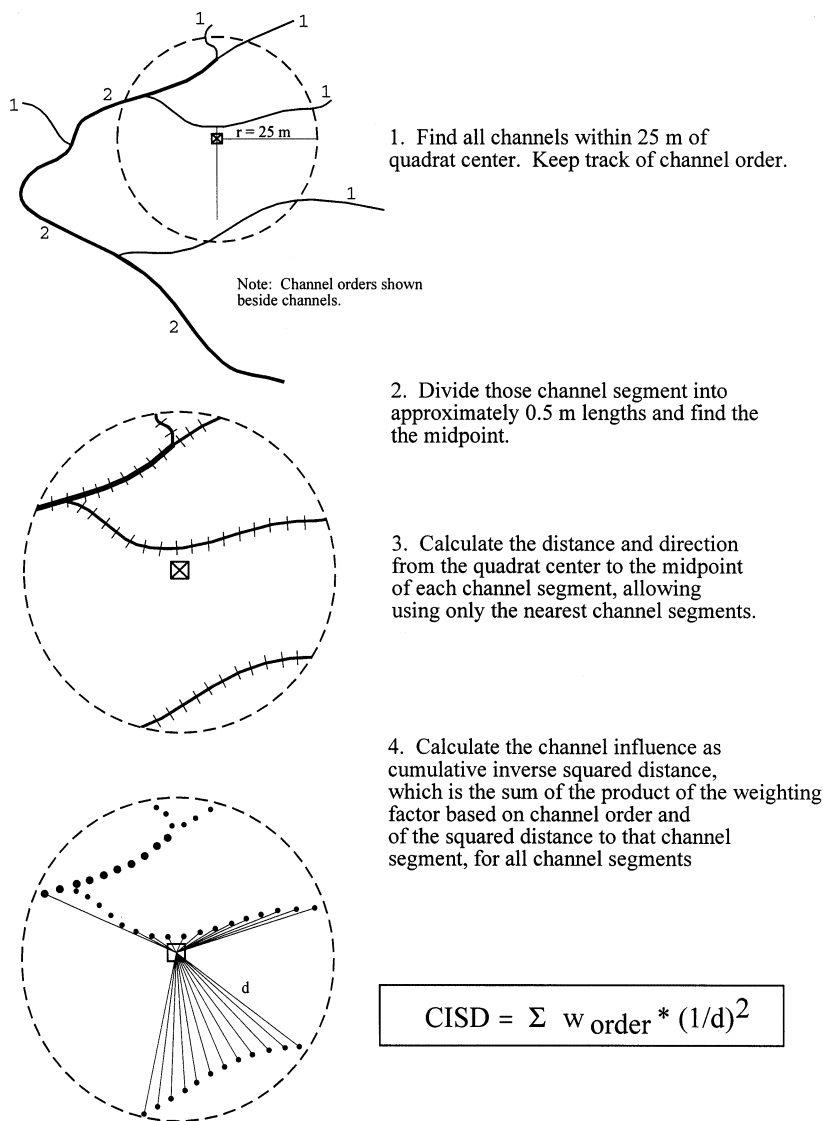


Fig. 2. Schematic diagram of the calculation method for CISD function.

where q is the particular location in the salt marsh, d is the distance from the location to a length of potentially influential channel, and w_o is the weighting factor based on channel size, as explained below.

The GIS algorithm used to calculate the CISD function is described schematically in Fig. 2. All channel lengths within a 25 meter radius of the location, q , were found. These channel lengths were divided into 0.5 m segmets. The distance (d)

and direction (rounded to the nearest degree) from q to the mid-point of each channel segment were calculated. From q , the algorithm searched through 360, 1° angles, taking only the closest channel mid point within each angle, to calculate CISD.

Weights were based on channel order and increased by one order of magnitude for each change in channel order (except second order channels) as shown in Table 1. Both channel size

Table 1

Assignment of channel order weighting factors is based on the approximate order of magnitude of channel cross-sectional area and maximum discharge measurements

Channel order	Assigned weight	Mean channel cross-sectional area (m ²) ^a	Estimated maximum discharge rate (m ³ s ⁻¹) ^b
1	0.1	0.06	0–0.03
2	0.5	0.14	0–0.03
3	1	1.12	0.04
4	10	2.35	0.29–0.82
5	100	33	4.5–17.8
6	1000	385	93.5–186.9

^a From Sanderson et al. (2000).

^b From Leopold et al. (1993).

(based on cross-section area from Sanderson et al., 2000) and estimated channel discharge (Leopold et al., 1993) vary by approximately one order of magnitude with each change in channel order and were used to calibrate the model weights.

The untransformed CISC function was calculated at a 0.5 m grid cell resolution for all cells within 10 m of any tidal channel and at 1.0 m grid cell resolution for the remainder of the site to match the sampling interval used by Sanderson et al. (2000). The larger grid cell data were then resampled back to 0.5 m grid cell size to create a continuous grid of consistent size (Fig. 3).

The untransformed CISC values calculated for the study site varied over fourteen orders of magnitude, so they were log transformed and then grouped using a multiplier and a “floor function.” A floor function rounds a number down, toward zero, to the nearest multiple of significance. In this case, multiplying the log transformed value by three, then taking the floor, lumps all CISC values within approximately one-half of an order of magnitude (producing 32 transformed CISC values for this study site). A constant was added to make all values positive. The final form of the transformed function was

$$= \text{floor}\{\lceil \log_{10}(\omega_0^* d^{-2}) \rceil * 3\} + 14. \quad (2)$$

The sensitivity of the model results to the formulation of this function is considered below. Through the remainder of the paper, references to “CISC” will indicate only the transformed function.

2.3. Calculation of species probability distributions as a function of CISC

Initially the CISC function was calculated for all the locations where we made plant cover observations (Sanderson et al., 2000). The probability of given species presence at a given CISC value was determined by counting the number of locations where that species occurred at a that

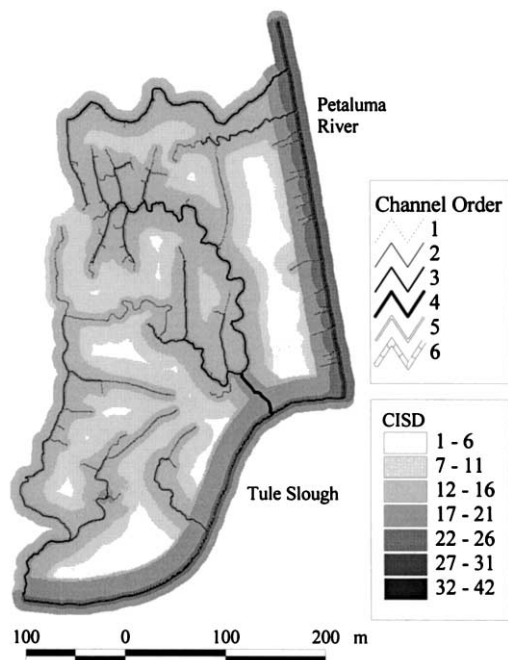


Fig. 3. Map of CISC values calculated for a portion of the tidal channel network at Petaluma Marsh, Sonoma County, CA, USA. The tidal channels were overlaid the CISC map.

CISD value, and dividing by the total number of locations with that CISD value. (For example, *Salicornia* was observed at 17–23 locations with CISD equal to 24, so the probability of presence was 0.74 for CISD = 24).

Because of the distinct clumping of species observed by Sanderson et al. (2000), we also calculated the probability of species presence, given presence of that same species in an adjacent quadrat (hereafter referred to as the “probability of aggregation”). These conditional probabilities were calculated as the product of the probability that a species occurs at a given location (the probability of presence, as above) and the probability that two adjacent quadrats both had that same species (Sokal and Rohlf, 1995). The probability of adjacent quadrats having the same species was calculated by counting the number of times two adjacent quadrats had the same species and dividing by the total number of quadrats at each CISD value.

Finally, for CISD values where a species was present, we also calculated the mean and standard deviation of the species cover to use in predicting species cover. These values were approximately normally distributed for all species as determined by the Shapiro-Wilk test (Shapiro and Wilk, 1965).

2.4. Stochastic predictions of species distributions

Species distributions at the study site were stochastically determined in a two step process. First for each species in each grid cell, the presence or absence was predicted by drawing a random number from a uniform distribution and comparing to the probability of presence calculated from the field data. Second for those cells where presence was indicated, adjacent cells were tested again using the probability of aggregation, up to three times for each species to match the observed patch size. This aggregation step simulated the observed patchiness of species distributions in the transect data.

Prediction of cover for cells where presence was indicated was conducted by drawing a normal random variate for that cell, with mean and

standard deviation for species cover as calculated previously. Total cover in a given cell was limited to 150% (the maximum observed by Sanderson et al., 2000). Cover was predicted for each species in order of that species’ total abundance as observed by Sanderson et al. (2000) at the study site.

The model was implemented using a combination of macro programming in ARC/INFO (ESRI, Redlands, CA) and scripting in perl (Larry Wall, Mountain View, CA).

2.5. Verification of vegetation distributions

To evaluate the model results, a vegetation map developed by Sanderson et al. (1998) for a separate remote sensing study was available. The vegetation map and simulation result for each species were plotted and qualitatively compared to verify overall trends in species distribution. The total amount of cover predicted by the model and observed in the vegetation map for each species was calculated for the coincident area.

The two spatial datasets were also compared on a cell by cell basis using an error matrix (Janssen and van der Wel, 1994). Both the vegetation map and the simulation were resampled to 10 m grid cells and reclassified as having high (25–100%), low (>0–25%) or no (0%) cover for a given species. These resampled datasets were used to calculate an error matrix and derive the percentage of correctly classified grid cells for each species.

2.6. Sensitivity analysis

Three aspects of the formulation of the CISD function were evaluated through sensitivity analysis: the decay of channel influence with distance, the exponential weighting scheme and the grouping factor. In each case the CISD values were calculated for the transect locations from Sanderson et al. (2000) and new species probabilities calculated to evaluate the sensitivity of species distributions to CISD.

3. Results

3.1. The spatial distribution of CISD

The map of the channel influence function (CISD) values across the study site is shown in Fig. 3. As determined by the formulation of the function, CISD is largest near the largest channels (higher channel orders), and progressively smaller with smaller channels (lower channel orders). Within a given channel order, CISD also declines exponentially with distance. Close examination of the CISD map indicates that CISD is not a simple buffer function around the channel networks; rather it takes into account the geometry of the network, generating higher CISD values inside channel bends than outside them, and allowing nearby larger channels to influence vegetation on the banks of smaller channels, even if the large channel is farther away. The generated values provide a complete estimate of the hydrologic influences arising from local channels across the modelled extent.

3.2. Species probability distributions with CISD

The probability distributions in Fig. 4 show where all species occur along the generalized channel influence (CISD) gradient. To facilitate interpretation of species presence values against CISD, representative axes showing distance from hypothetical isolated and straight channels of first, third and sixth order are shown parallel to the ordinate axis of Fig. 4.

The distribution of species distributions with CISD conforms surprisingly well to the observations plant ecologists have made in northern California salt marshes. *Salicornia* is the salt marsh dominant in these salt marshes, with up to 100% cover in some locations (Josselyn, 1983). In this case, the probability of *Salicornia* presence varies between 0.9 and 1.0 for locations with CISD 8–23, then declines sharply with increasing channel influence. The sharp ecotone between *Salicornia* and *Spartina*, has been noted by plant ecologists for over 50 yr (Purer, 1942; Hinde, 1954; Chapman, 1960) and was extensively studied in a series of papers by (Mahall and Park

1976a,b,c) and by Brenchley-Jackson (1992). The abrupt nature of this ecotone is indicated by the steep slope of the probability curves at CISD values 23–26. At CISD values above 28, which only occur along the largest channels, *Spartina* dominates.

The distribution of *Scirpus*, studied by Ustin et al. (1982) and Whitlow (1982), occurs between CISD 18–28, with highest probabilities occurring at CISD 22–24, a little farther from the large channels than the *Spartina*/*Salicornia* ecotone, as observed in the field. Because *Scirpus* grows in irregular patches, not a continuous swath, its highest probability of presence is approximately 75%.

Several species, including *Grindelia*, *Frankenia*, *Lepidium* and *Distichlis*, cluster at CISD values 16–25, peaking in the range of 18–22. These overlapping distributions correspond to associations of species reported by many authors growing along side the banks of third and fourth order channels (De Groot, 1927; Purer, 1942; Marshall, 1948; Hinde, 1954; Chapman, 1960; Vogl, 1966; Redfield, 1972; Balling and Resh, 1983; Collins and Resh, 1985; Adam, 1990; Wayne and Parker, 1994, 1996; Sanderson et al., 1998). The distribution of *Frankenia* broadly overlaps the other species in the mid-CISD range, but also extends to lower CISD values (11–25). *Jaumea* similarly overlaps other species in the mid range, but also has an important presence at approximately 30% probability in the lower CISD values. Sanderson et al. (2000) reported an idiosyncratic distribution of *Jaumea* within the *Salicornia* “background” that may be associated with slight depressions in the marsh plain, which may be associated with mycorrhizal associates (Brown and Bledsoe, 1996).

3.3. Simulation of species distributions

Simulated distributions of individual species compare favorably qualitatively and quantitatively with known distributions of these species from the literature and the vegetation map compiled by Sanderson et al. (1998). The overall cover predicted by the simulation for each species compares favorably to the overall cover estimates

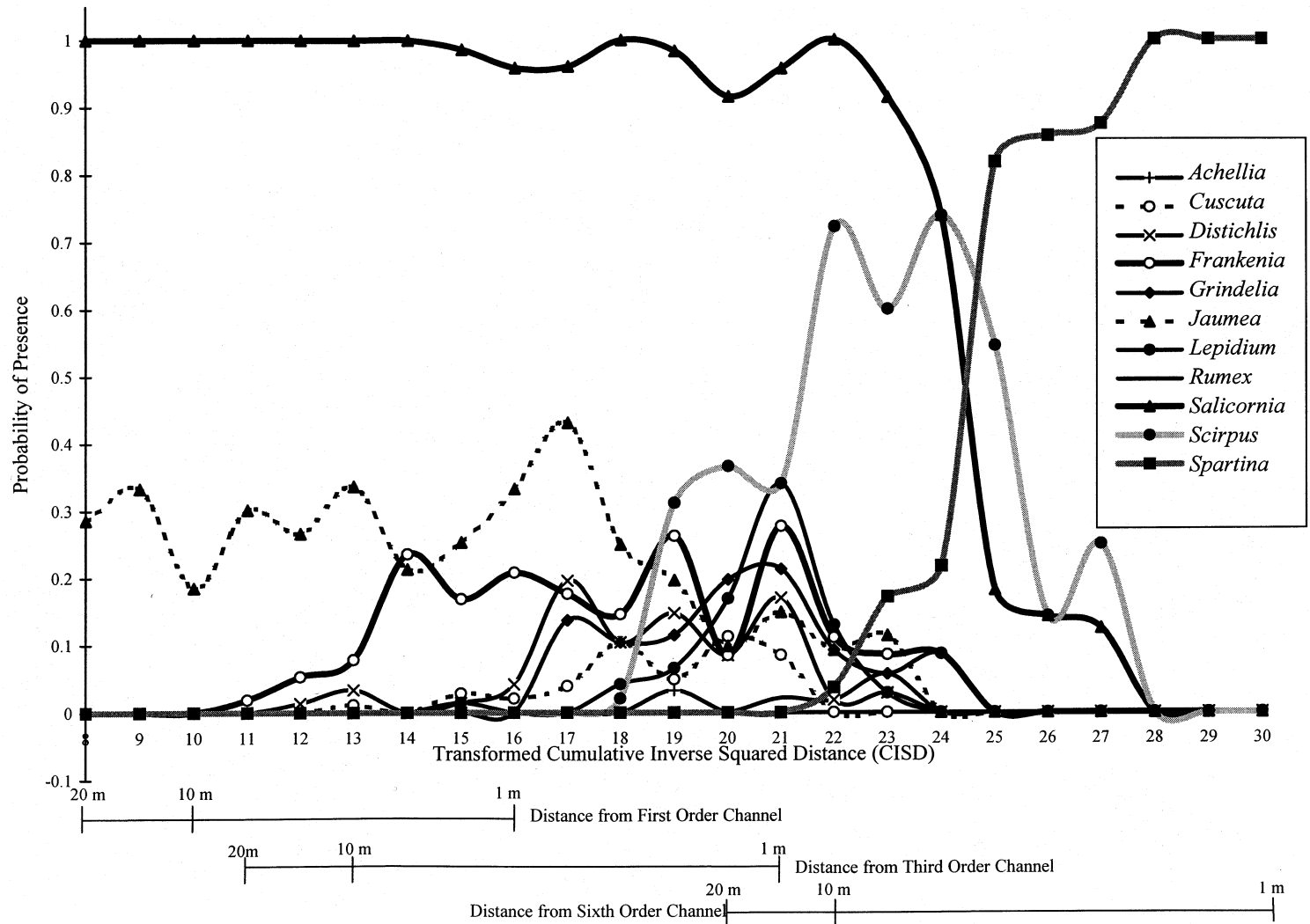


Fig. 4. Distribution of probabilities of presence for twelve salt marsh plant species with CISD. Additional ordinate axes show the calculated distances from isolated straight channels of first, third and sixth order for comparison to CISD values.

Table 2

Comparison of the overall amount of cover recorded in the vegetation map of Sanderson et al. (1998) and the simulated model distributions for those species based on the coincident area

Species	Total cover from vegetation map (%) ^a	Total cover from simulations (%) ^b
<i>Salicornia virginica</i>	82.95	77.24
<i>Spartina foliosa</i>	1.45	0.69
<i>Scirpus robustus</i>	11.07	9.38
<i>Frankenia salina</i>	1.26	2.40
<i>Grindelia stricta</i>	0.67	0.39
<i>Lepidium latifolium</i>	0.43	0.16
<i>Jaumea carnosa</i>	0.20	2.34
<i>Rumex crispus</i>	0.03	0.02
<i>Achillea millefolium</i>	0.02	0.02
<i>Cuscuta salina</i>	<0.01	0.03
<i>Distichlis spicata</i>	not observed	0.20
<i>Atriplex patula</i>	not observed	<0.01

^a Sanderson et al. (1998).

^b This study.

made during the vegetation mapping for most species (Table 2) – all discrepancies are less than 6%. The discrepancies that do exist may be attributed to the coarse scale and limited extent of the vegetation map (e.g., *Spartina*, *Jaumea*) or poor fits to the model (e.g., *Scirpus*, see below).

On a cell by cell basis, using the resampled 10 m grid cells, there are very high percentages of correctly predicted cells for most species (Table 3), though this test is a somewhat weak test because of the coarse spatial and classification resolution used. Six of the ten modelled species are correctly predicted for more than 90% of the cells; the lowest prediction is for *Scirpus* at approximately 73%.

On a qualitative basis, the simulations also appear to do well for most species. *Salicornia* (Fig. 5a) is shown nearly ubiquitously across the marsh, except where it intermixes with *Scirpus*, and then declines sharply beside the river, in the *Spartina* zone. It is also slightly less frequent along the banks of intermediate sized channels (third and fourth order), leaving patches where *Grindelia* and *Frankenia* occur.

Spartina (Fig. 5b) lines the banks of the largest channels, particularly in a continuous belt along the Petaluma River (a sixth order channel) approximately 10 m wide, and in a narrower band (~ 3 m wide) along Tule Slough (fifth order). Small patches, in the field associated with slump blocks lower than the marsh plain, are observed within the central, fourth order channel.

Of all the species, the model prediction for *Scirpus* is the least representative (Fig. 5c). The model predicts *Scirpus* in a nearly continuous band, approximately 20 m wide along the Petaluma River, and with lesser width along Tule Slough and the fourth order channel. In contrast the vegetation map shows *Scirpus* growing in primarily one large patch, extending nearly 40 m inland, along the Petaluma River, with only low amounts of cover along Tule Slough.

The aboveground distribution of *Scirpus* is known to vary interannually with changes in soil salinity due to changes in precipitation amount

Table 3

Accuracy assessments derived from cell by cell comparisons of vegetation map of Sanderson et al. (1998) and the simulated model distributions for those species for the coincident area

Species ^a	Percentage cells correctly predicted compared to vegetation map (%)
<i>Salicornia virginica</i>	92.61
<i>Spartina foliosa</i>	94.83
<i>Scirpus robustus</i>	72.66
<i>Frankenia salina</i>	77.34
<i>Grindelia stricta</i>	88.67
<i>Lepidium latifolium</i>	94.83
<i>Jaumea carnosa</i>	79.06
<i>Rumex crispus</i>	98.77
<i>Achillea millefolium</i>	99.26
<i>Cuscuta salina</i>	99.51

^a No comparison data available for *D. spicata* and *A. patula*.

Note: Both the map and the simulations were resampled to coincident 10 m grid cells and aggregated into zero, low (0–25%), and high (25–100%) classes. An error matrix was calculated, but not shown, in order to calculate accuracy assessment measures.

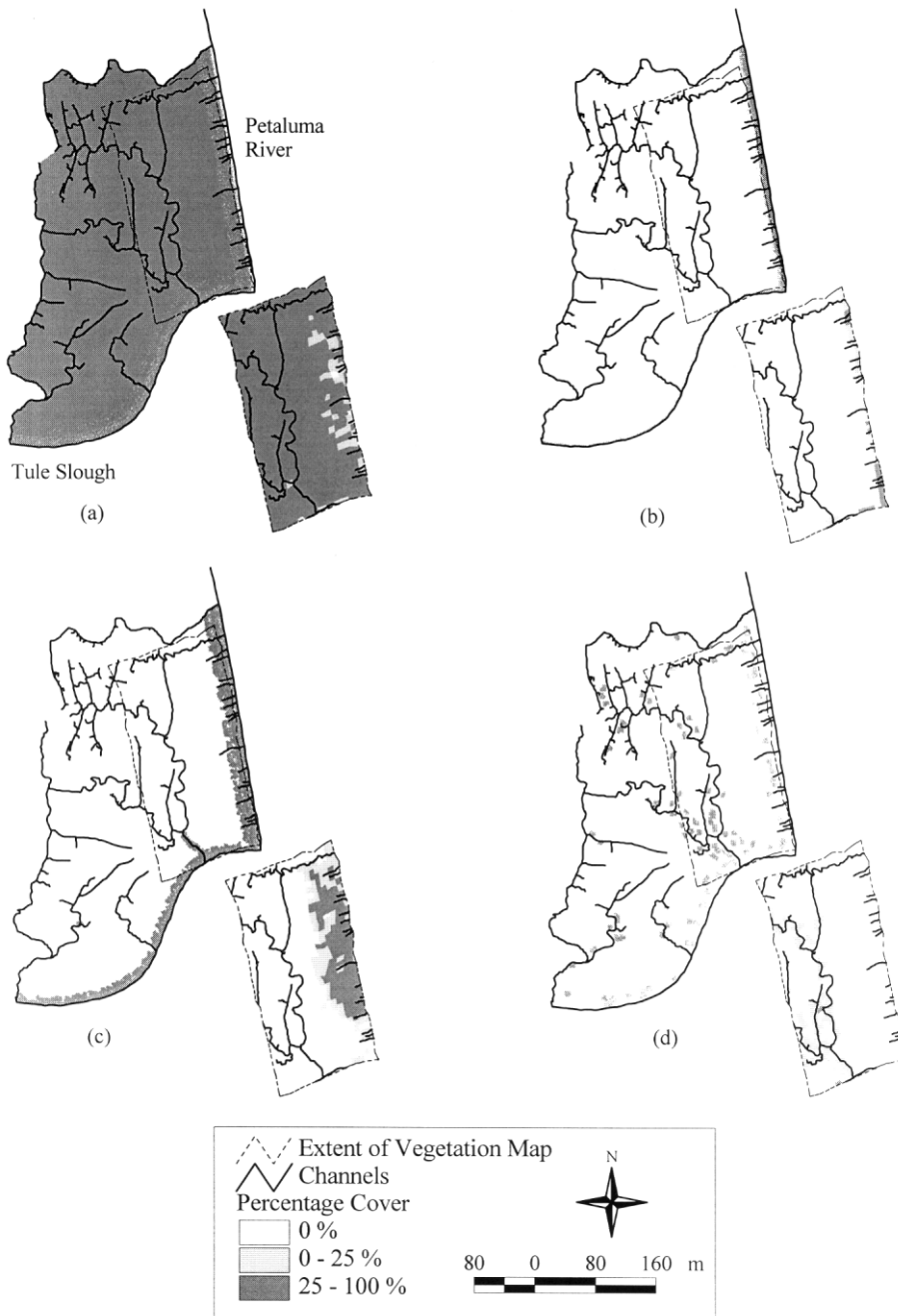


Fig. 5. Simulated spatial distributions for four representative plant species at Petaluma Marsh: (a) *Salicornia virginica*, (b) *Spartina foliosa*, (c) *Scirpus robustus*, (d) *Frankenia salina*. *Grindelia stricta*, *Lepidium latifolium*, *Jaumea carnosa*, *Rumex crispus*, *Cuscuta salina*, *Distichlis spicata*, *Achillea borealis* and *Atriplex patula*, were also simulated, but are not shown. Vegetation distributions measured by Sanderson et al. (1998) in 1994 over a subset of the study site are shown offset and to the right of the simulated distributions. The coincident area is indicated with dotted lines on the simulated distributions.

which release or inhibit shoot production from rhizomes (Ustin et al., 1982). The years since 1994 (a dry year, when the vegetation map was prepared) have had normal or higher precipitation, and accordingly, the authors have noted that this patch has dramatically expanded its distribution extending along both the Petaluma River and Tule Slough, although not to the extent shown in the simulation results. Since the model does not include dynamic climatic changes and is based on data only from a single season, it is unable to reflect these interannual changes in distribution.

The model simulates the levee flora association of plants associated with mid-sized channels. *Frankenia* (Fig. 5d) is indicative of this vegetation association and is shown in close association with the intermediate sized channels of third and fourth order. *Frankenia* was also predicted by the model to present with low cover within the *Scirpus* zone, as were several of the other typical levee flora species. These species were for the most part not observed in this area, except along the small first order channels which drain into the Petaluma River.

Distributions of *Lepidium*, *Grindelia*, *Jaumea*, *Achillea*, *Rumex* and *Cuscuta* were all also adequately predicted, though these results are not shown. In each case, the distributions generally fit the pattern observed in the vegetation map, though not exactly (Table 3). *Lepidium*, an exotic invasive, was predicted in association with the fifth and sixth order channels and has been observed to expand its distribution in recent years along with *Scirpus* (Sanderson, pers. obs).

Overall the model appears to do an adequate, but not perfect, job of predicting the spatial distributions of both major and minor wetland species. It predicts ecotonal boundaries in approximately the correct locations and it recreates associations of species also observed in the field. Although not an ideal fit, these results support the conclusion that channel size and distribution are important general factors structuring the distribution of plant species at Petaluma Marsh.

3.4. Sensitivity analysis

Models of this kind are particularly sensitive to

the details of the model formulation. We independently tested the order of the decay function, reformulated the weighting scheme according to a linear rather than exponential scale, and considered the effect of changing the grouping value used with the floor function. In each case, the result was unrealistic predictions of species distributions which matched poorly to the independently derived field map. For example, using d^{-1} resulted in a much less dramatic drop in CISD with distance resulting in species predictions which covered much larger areas than observed. Using a linear scale weighting scheme lessened the importance of difference in channel order in the CISD calculation and thus contracts the total range of CISD values, so that calculating species probability distributions against the smaller range of CISD values failed to resolve species separately along the channel influence function. Species predictions resulted in co-occurrence of sets of species which were not observed in the field. Finally, by examining the sensitivity of the model to the grouping multiplier, we found that higher multipliers tended to lump the probabilities, resulting in poor discrimination between species; lower values tended to dissect the probability distributions too finely, creating noisy, irregular distributions of species which were more random than observed in the vegetation map. Thus, the results obtained are highly dependent on the model form, reinforcing the importance of the exponential-decay function and the log-linear weighting scheme which themselves reflect observed hydrodynamic structures.

4. Discussion

The documentation of tidal control of tidal marsh vegetation was started several decades ago (Purer, 1942; Hinde, 1954; Mahall and Park, 1976a,b; Zedler and Beare, 1986) and attention has long since turned to the influence of biological interactions (Bertness and Ellison, 1987; Pennings and Callaway, 1992; Hacker and Bertness, 1995, 1999) and more recently, to interactions between physical and biological factors (Bertness 1988; Bertness and Hacker, 1994; Brewer et al., 1998; Dijkema, 1998; Yang, 1998). Nevertheless, the

ability of this model to correctly predict the Petaluma Marsh data is remarkable, given that the predictions are based on a randomized three-stage estimation of site CISC value, species presence and cover value, independent of detailed understanding of plant physiological or ecological processes. All three steps are also independent of the vegetation map to which the model is compared, which inspires greater confidence in the conclusion that the vegetation of Petaluma Marsh is influenced by tidal flows. These results emphasize that marsh vegetative development is an ecosystem process (Teal, 1962; Redfield, 1972; Hopkinson and Day, 1986; Wiegert, 1986; Schlesinger, 1991; Newman et al., 1996).

The results of the sensitivity analysis revealed that the strong correspondence between field data and model predictions are sensitive to model structure, particularly the use of logarithmic scaling and exponential decay with distance. These results reflect the importance of logarithmic scaling of stream order, in parallel to tidal volume and the interaction between soil and water in the marsh surface and subsurface. These findings reinforce our conclusion that tidal hydrology must be a prime consideration in marsh vegetative development.

In contrast, the most important caveat is that the strong association between channel influence and vegetation pattern does not measure system hydrology alone, but rather the entire suite of influences related to tidal volume and influx at any site in the marsh. These influences are truly diverse and sometimes diametrically opposed (e.g., frequent tidal influx restoring soil moisture, infrequent tidal influx promoting higher soil salinities) or compounded in complex ways (e.g., tidal flows as agents of sediment transport, soil waterlogging as a function of vegetation and water movement above and through the soil). None of these more detailed causal pathways, however, changes our main conclusion.

While CISC evidently works quite well for Petaluma Marsh, we have not yet tested its ability to predict vegetation structure in other tidal marshes. We have been collecting data for two other marshes within the northern San Fran-

cisco Bay–Delta complex (Fagan Slough, Napa County, and Rush Ranch area of Suisun Marsh, Solano County) that generally have lower aqueous salinities and therefore should exhibit lower soil salinities in their upper reaches (Culbertson et al., 2000). The vegetation in Petaluma Marsh is demonstrably more clearly zoned and less diverse than that of the other marshes (Zhang et al., 1997), so it is possible higher salinities or other factors produce sharper gradients in the vegetation. Similar patterns have been described in Tijuana Estuary and other Southern California estuaries, where precipitation is lower and estuaries are dominated through most of the year by the influx of seawater (Zedler et al., 1996).

The potential value of our findings lies in two areas. Our results do not advance the case for particular causal pathways, such as facilitation by aeration (Hacker and Bertness, 1999), or add any new factors influencing marsh vegetation to our understanding. What they suggest instead is that these casual processes should be understood in the context of marsh hydrology – the distribution of channels and channel cross-section – which drives the larger scale marsh structure and likely marsh function. Thus our model stresses the need to restore tidal flushing through channel networks when creating or restoring salt marsh ecosystems (Desmond et al., 2000). This is a general principle which needs further elaboration, and is receiving it in experimental tests of hydrologic restoration in experimental blocks in Tijuana Estuary (Callaway et al., 1997).

The CISC model might find use in systems where it is desirable to establish a system of precursor channels to direct flow within the developing marsh. Alternative drainage configurations could be tested in advance by using GIS and CISC to predict future vegetation pattern and establish monitoring protocols to follow marsh development. These applications underscore the utility of expressing natural phenomena as functions of distance. By expressing our model through distance based relationships, we can bring to bear significant analytical capabili-

ties and processing power of modern GIS systems, which allow us to extrapolate patterns through two dimensional space. Used in either the more general sense of setting priorities or in the more specific sense of directing marsh restoration schemes, we expect that this research will help increase the success of restoration programs and enhance the conservation of these important ecosystems.

5. Conclusion

Tidal channel networks strongly structure salt marsh plant distributions in Petaluma Marsh, CA. A simple, empirical measure of channel influence, based on cumulative distance from the channel network and channel size, was generally able to predict the distributions of both major and minor plant species at one site. These results emphasize the importance of tidal channel networks in salt marsh ecosystem structure and function.

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